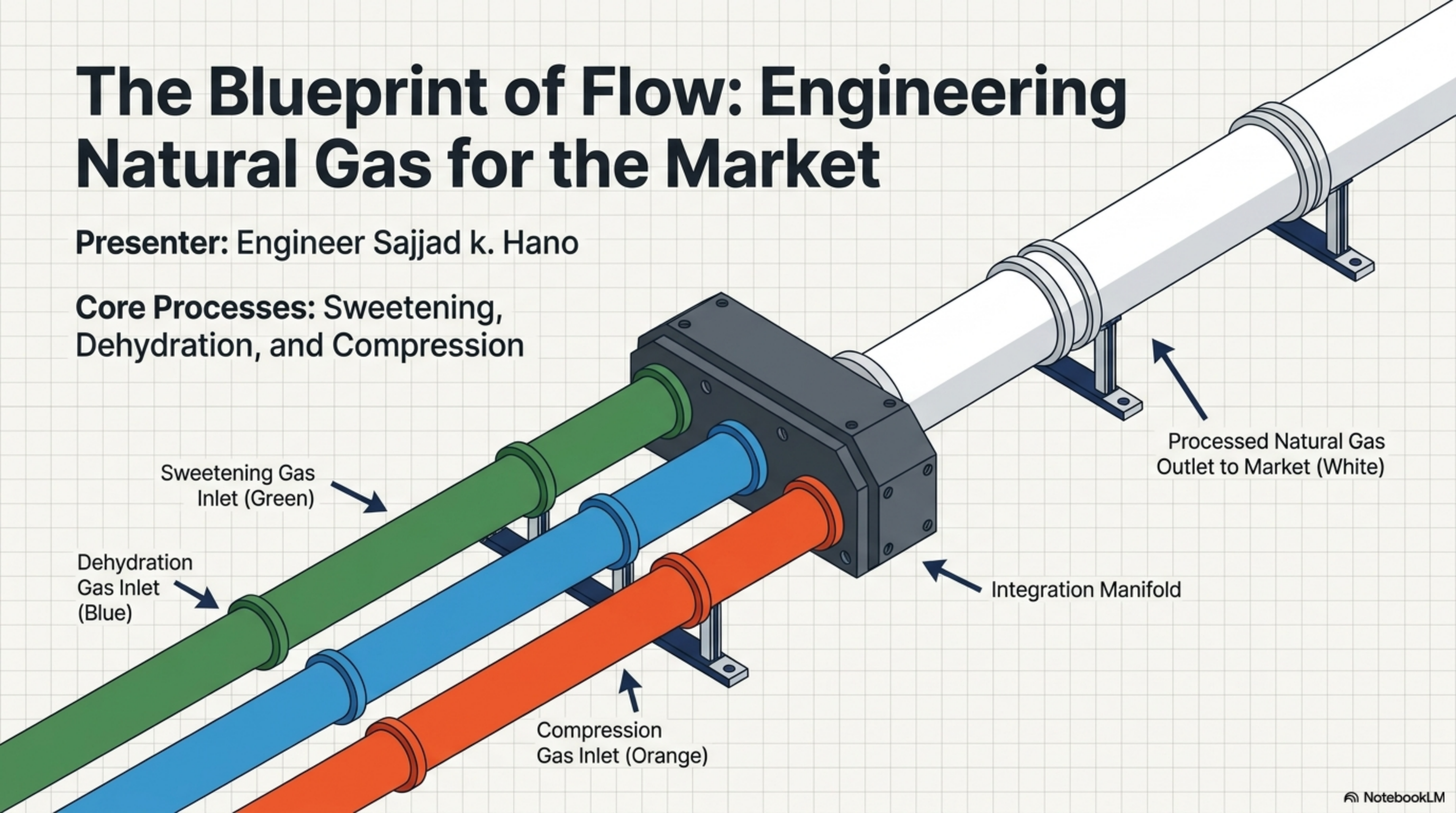


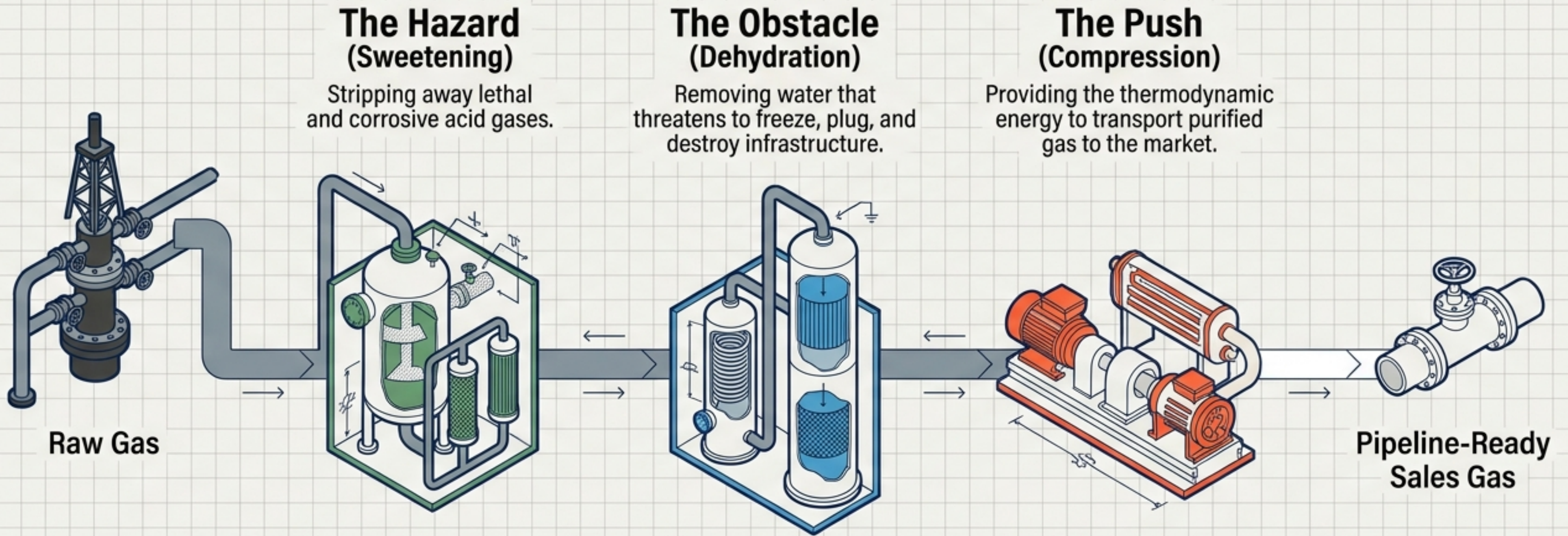
The Blueprint of Flow: Engineering Natural Gas for the Market

Presenter: Engineer Sajjad k. Hano

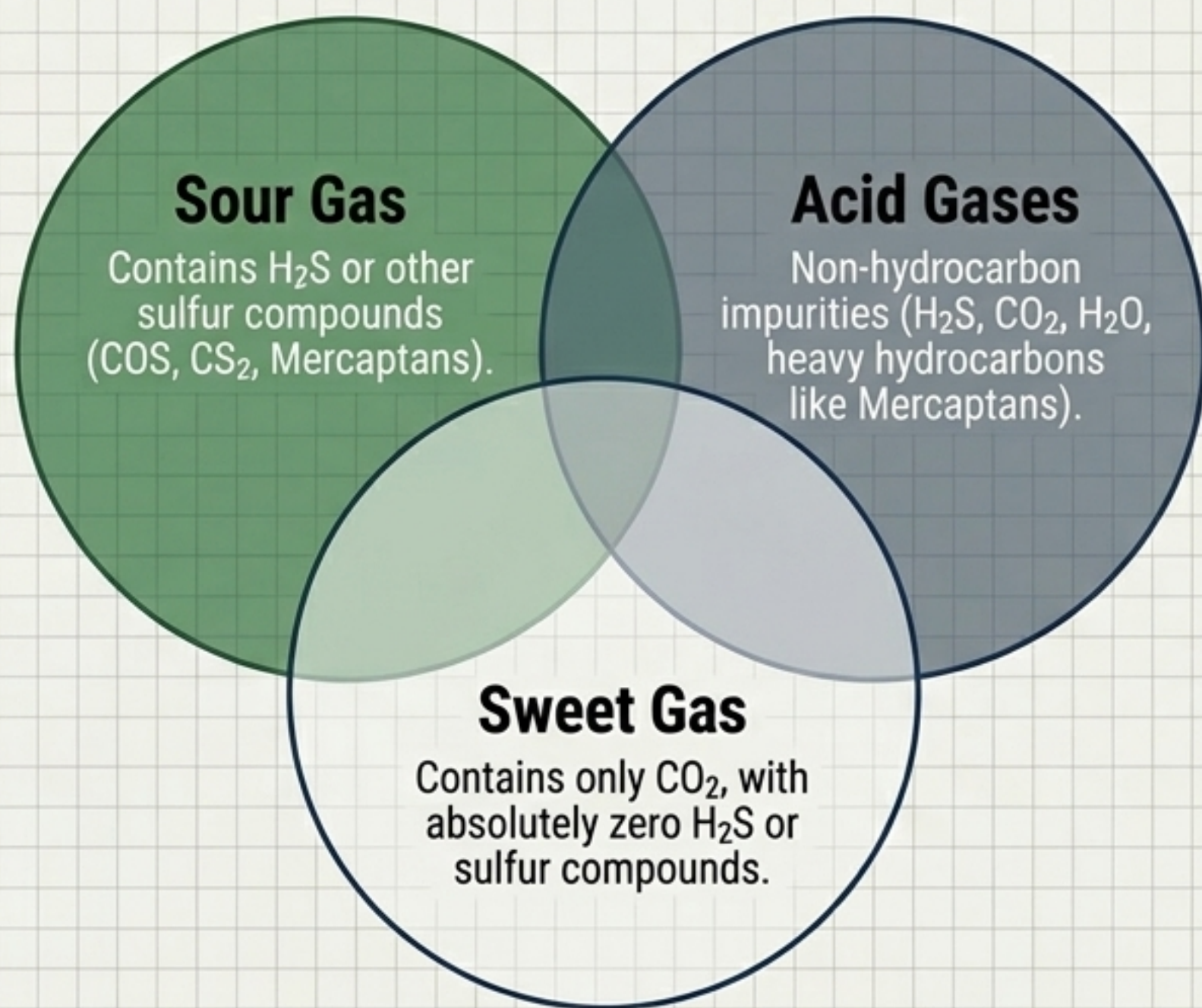
Core Processes: Sweetening, Dehydration, and Compression



Raw natural gas emerges as a hazardous, wet, and sluggish mixture



Categorizing impurities differentiates sour gas from sweet gas

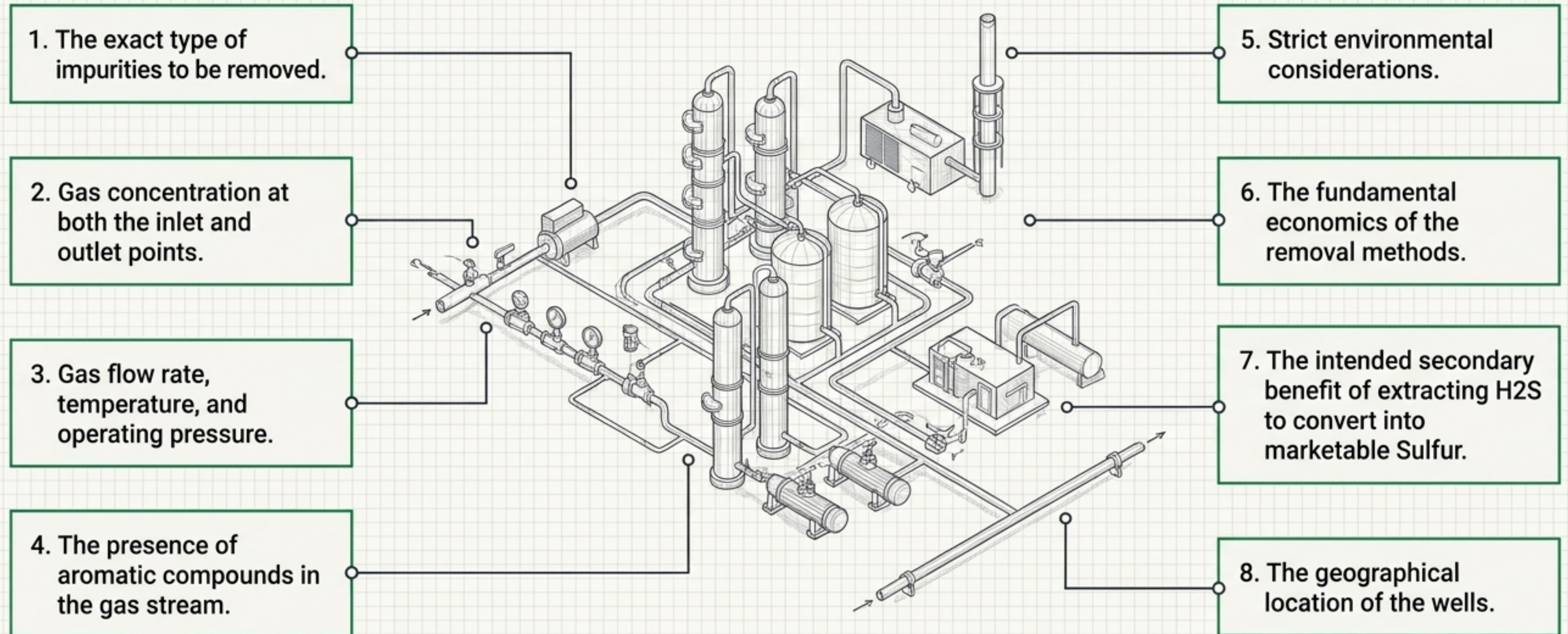


3 Reasons for Gas Sweetening

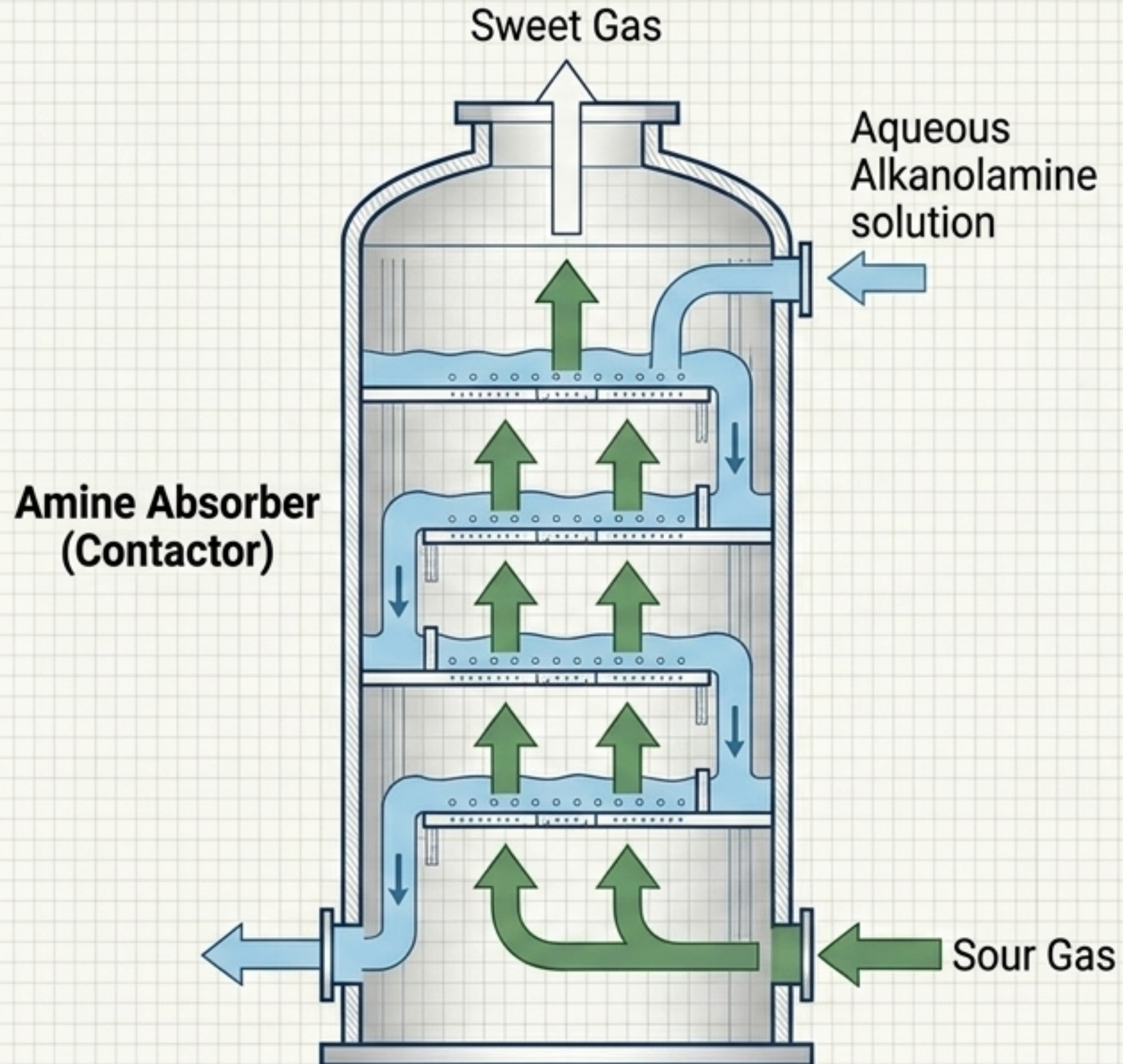
1. Mitigating immediate health hazards.
2. Meeting strict sales contracts.
3. Eliminating severe pipeline corrosion problems.

Eight engineering factors govern the selection of a gas sweetening process

System selection requires balancing chemical, operational, and economic variables:



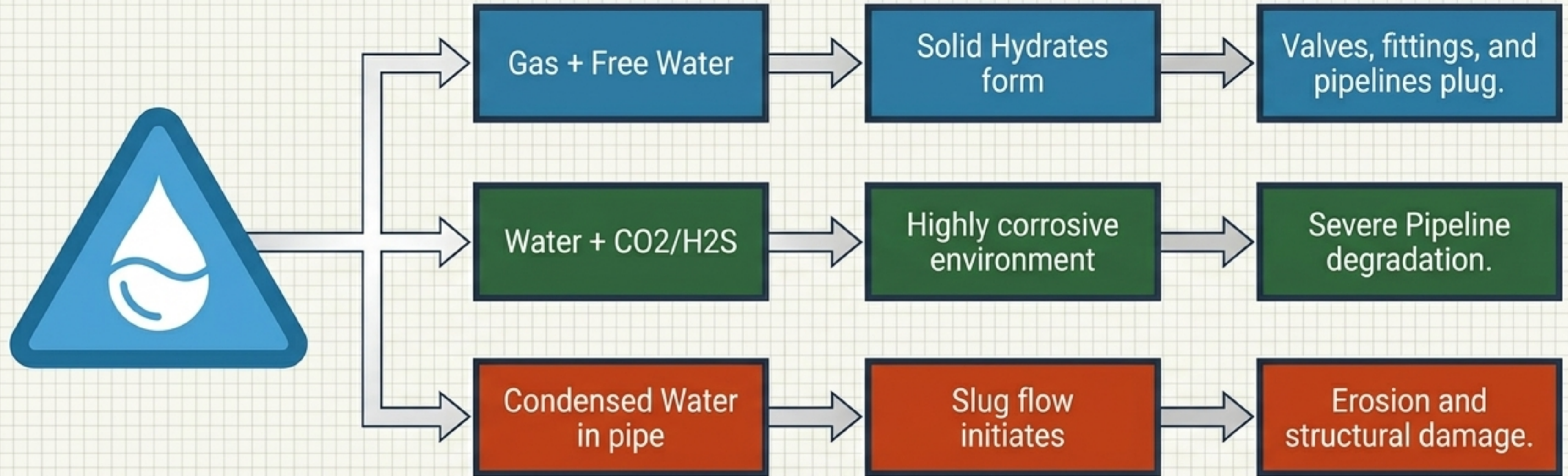
The Amine Process relies on counter-current absorption to strip acid gases



The Mechanism:	Gas bubbles counter-currently through liquids that have a special chemical attraction (affinity) for impurities.
The Solvents:	• MDEA (Methyldiethanolamine) • DEA (Diethanolamine) • MEA (Monoethanolamine)
Operating Conditions:	System efficiency is dictated by circulation rate, solvent concentration, acid gas loading, and the steam rate required for regeneration.

Free water threatens to freeze, plug, and destroy pipeline infrastructure

Risk Map



Economic Impact

Water vapor unnecessarily increases volume while decreasing the heating value of the gas.

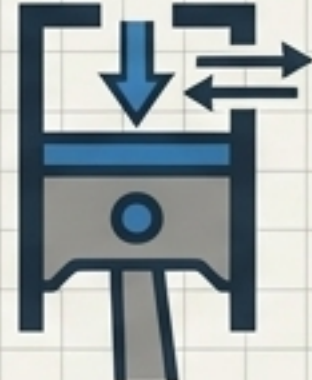
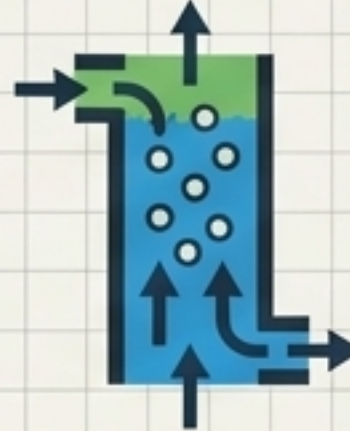
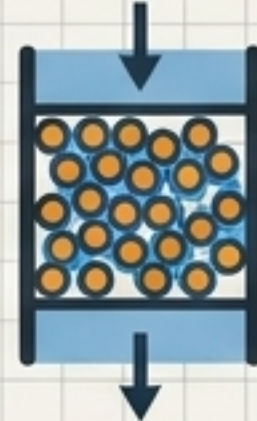
Operational Freezes

Dehydration is mandatory to prevent freeze-ups in cryogenic and refrigerated absorption plants.

The Contractual Limit

Sales gas contracts enforce a strict maximum water content of 7 lb H₂O / MMSCF (or a specific dew point).

Four primary dehydration categories dictate the removal of water vapor

1. Direct Cooling	2. Condensation	3. Absorption	4. Adsorption
			
Lowering temperature to force water out of the vapor phase.	Compression followed immediately by cooling.	(Glycol Dehydration) Gas bubbles counter-currently through liquid solvents with a special affinity for water. The industry standard to meet pipeline sales specs.	(Solid Desiccant) Gas flows through a bed of granular solids (Molecular Sieve, Silica gel, or Activated Alumina). Water is retained directly on the surface of the solid particles.

Note on Inhibition: Alternatively, chemical inhibitors can be added directly to condensed water to prevent hydrate formation without removing the water entirely.

Glycol selection depends on viscosity, vapor pressure, and hydrocarbon solubility

MEG (Monoethylene glycol / EG)

DEG (Diethylene glycol)

TEG (Triethylene glycol)

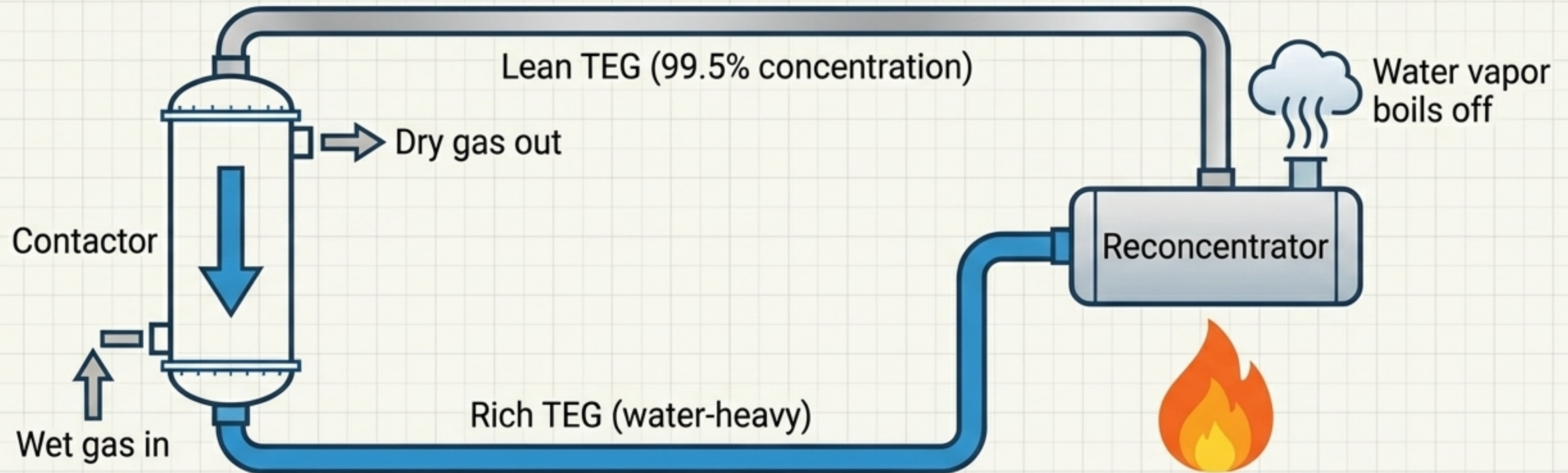
TREG (Tetraethylene glycol)

Ideal Solvent Properties

To be effective in the Contactor (Sorber), the glycol must be:

- ✓ Highly hygroscopic (water-attracting).
- ✓ Noncorrosive and non-volatile.
- ✓ Insoluble in liquid hydrocarbons.
- ✓ Nonreactive with hydrocarbons, CO₂, and sulfur compounds.
- ✓ Easily regenerated to high concentrations.

The TEG regeneration cycle maintains continuous water absorption



Design Criteria 1 (GWR)

2 to 6 gallons of TEG per lbm of H₂O removed is adequate for most requirements.

Design Criteria 2 (Lean TEG)

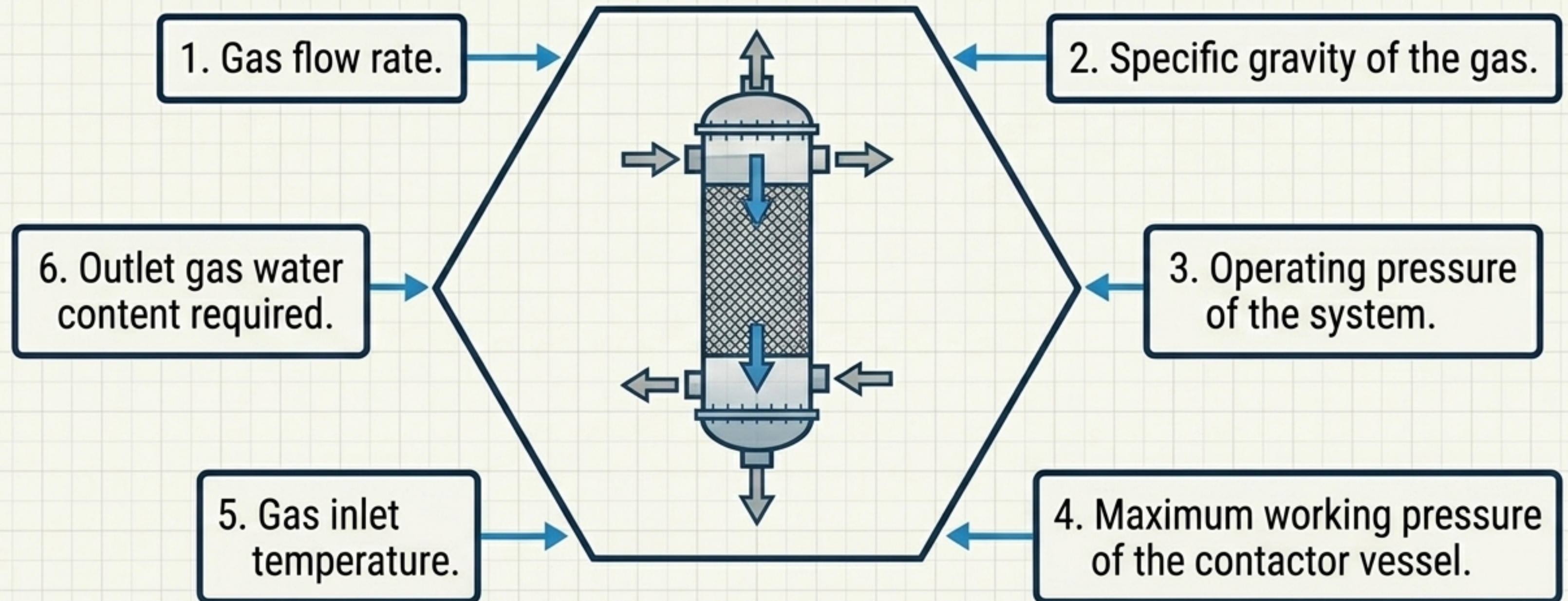
Most reconcntrators output 99% to 99.9% lean TEG. 99.5% is the standard utilized in most designs.

Absorption Yield

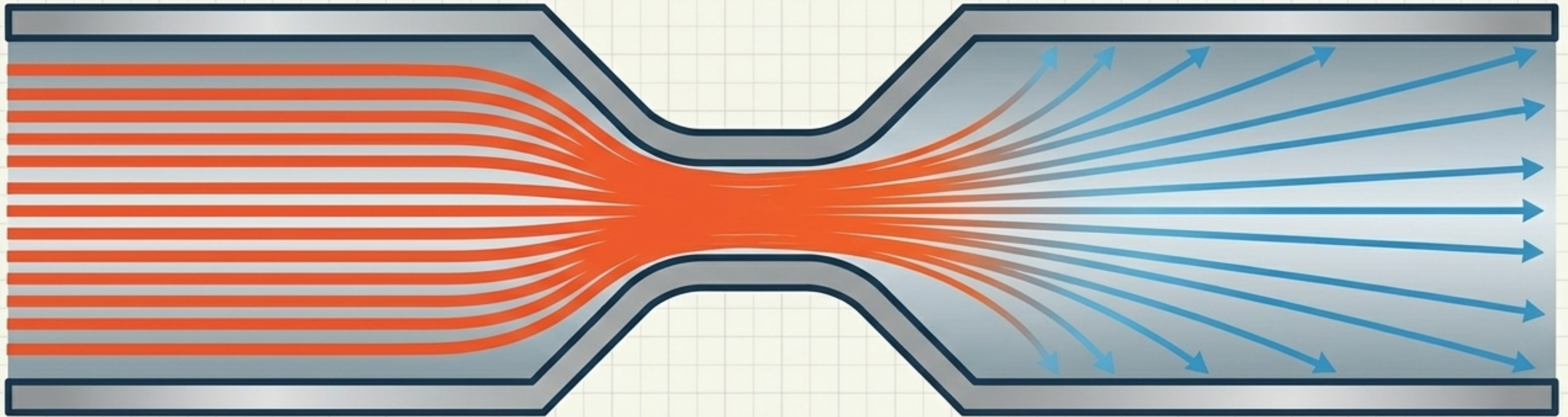
Total water removed depends on glycol concentration, circulation rate, and the number of contacts in the absorber.

Six precise variables dictate the sizing of TEG column contactors

Dehydration of Natural Gas with Triethylene Glycol in trays or packed columns requires specific operational data:



Expanding fluid through a choke triggers the Joule-Thomson temperature drop



The Joule-Thomson Effect

The specific temperature change (usually a drop) that occurs when a fluid expands through a restriction, causing an immediate decrease in pressure. Surface chokes (Positive or Adjustable) are engineered to manage this expansion.

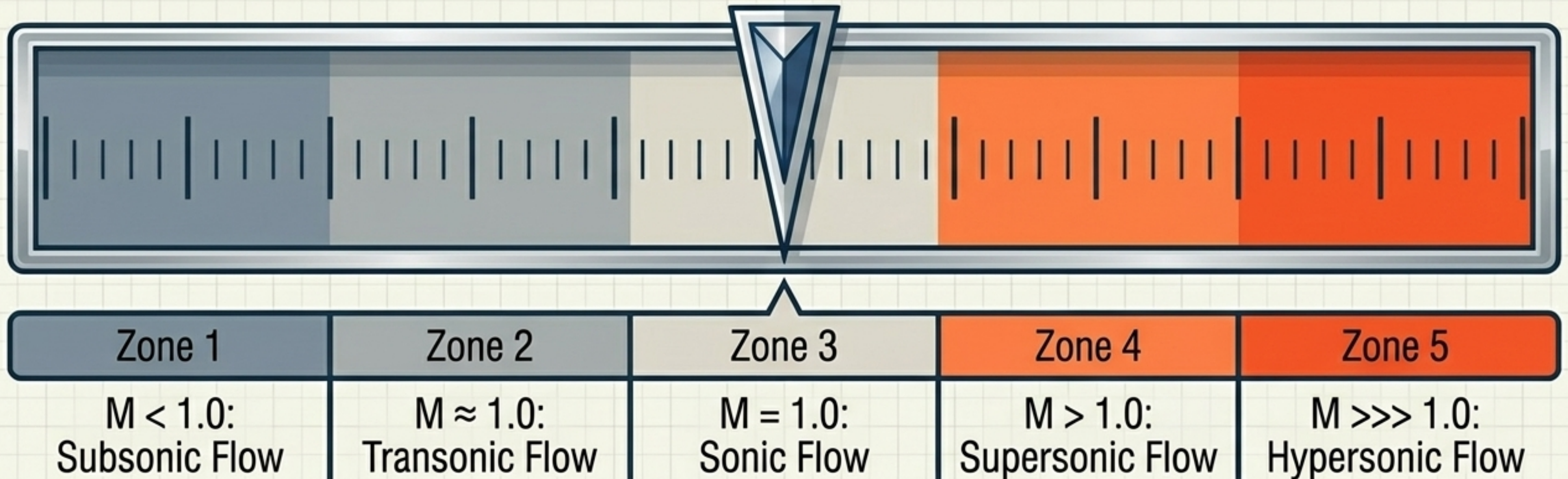
Specific Heat

The energy required to raise the temperature of a unit mass by one degree. Always follows the thermodynamic rule: $C_p > C_v$ [$k > 1.0$].

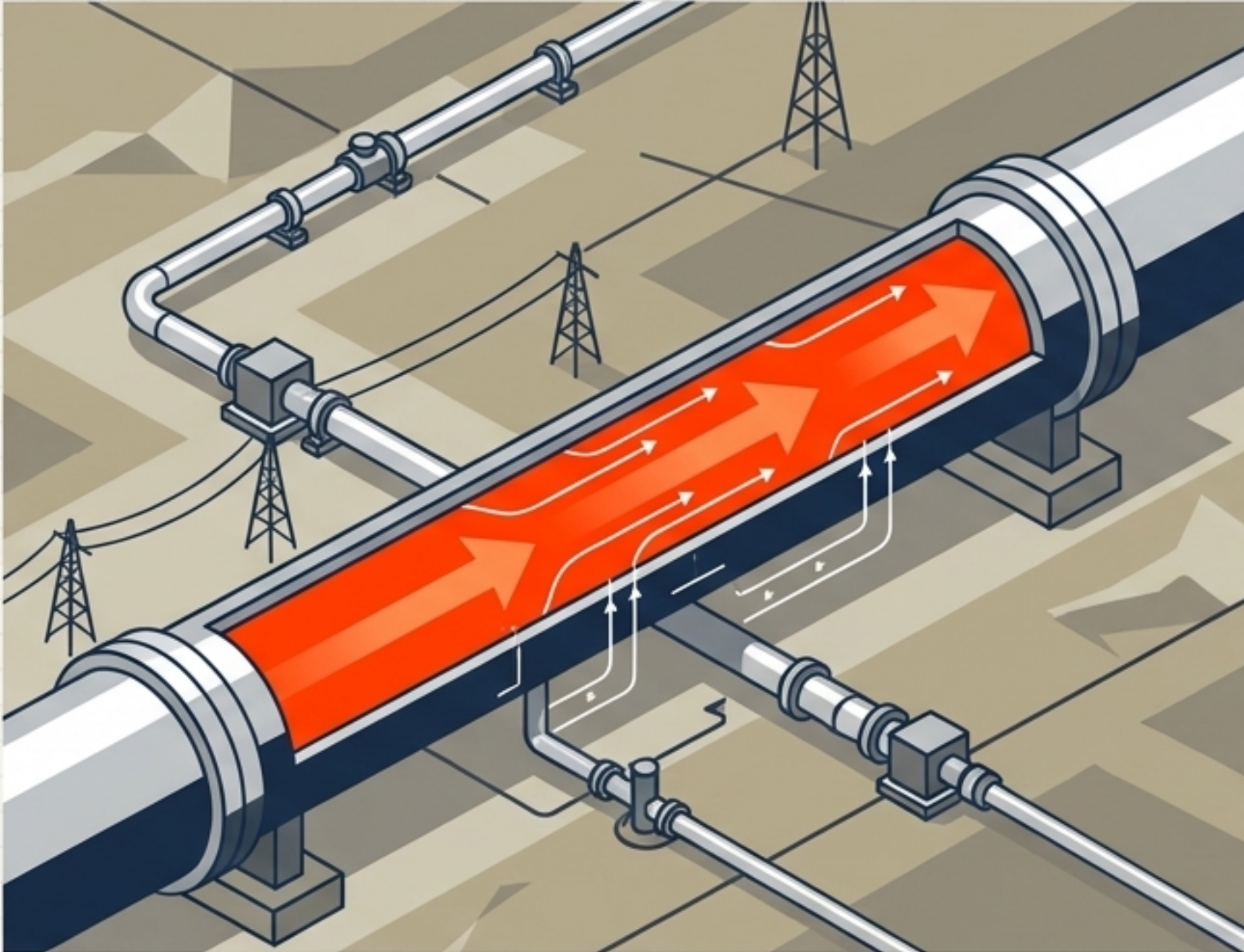
Mach Numbers index fluid velocity against the localized speed of sound

Sonic Velocity: The precise velocity at which an infinitesimally small pressure wave (or disturbance) propagates through a specific fluid medium.

Mach Number (M): The vital ratio between fluid velocity and sonic velocity in that same fluid.



Gas compression provides the kinetic energy required for transport and injection



Core Applications:

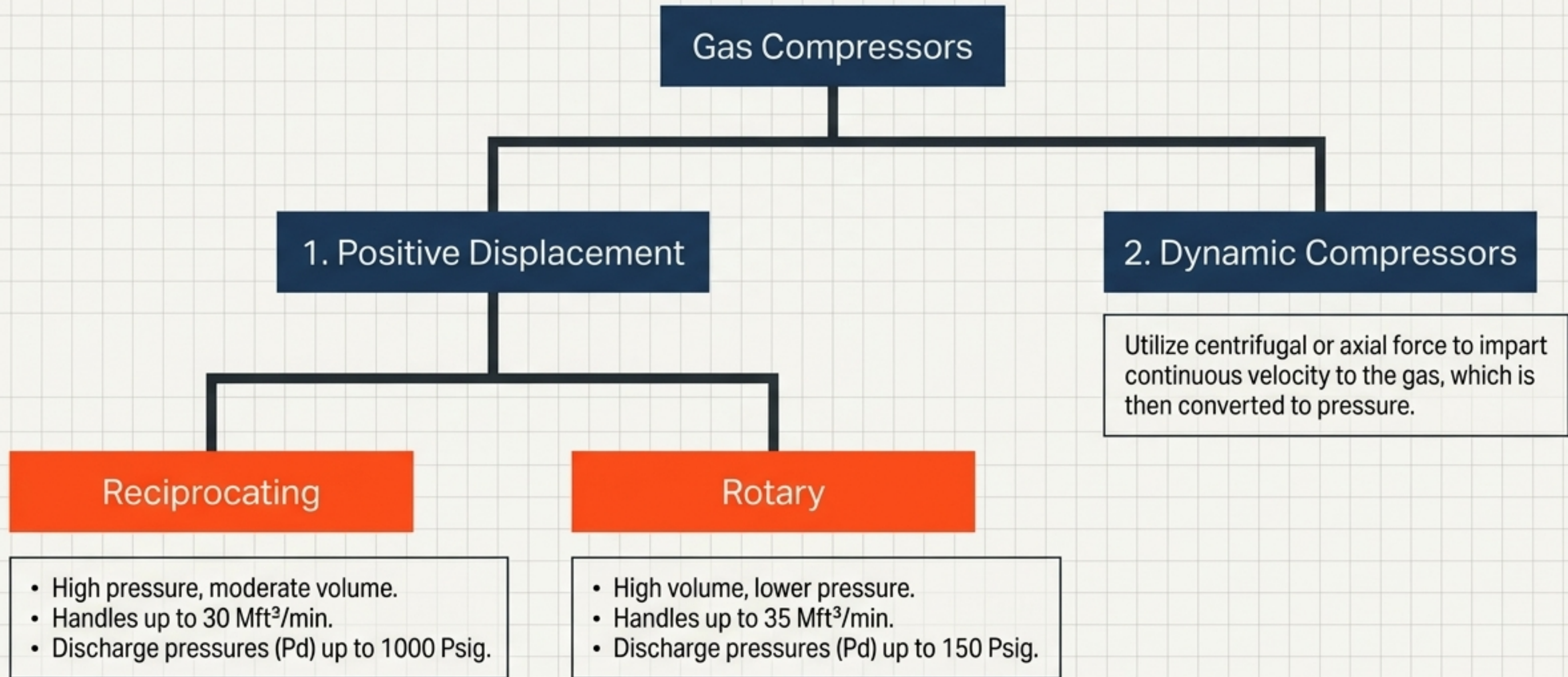
1. Overcoming pipeline friction for long-distance Gas Transportation.
2. Powering Gas Injection Projects for Gas Lift Operations.

Design & Price Factors:

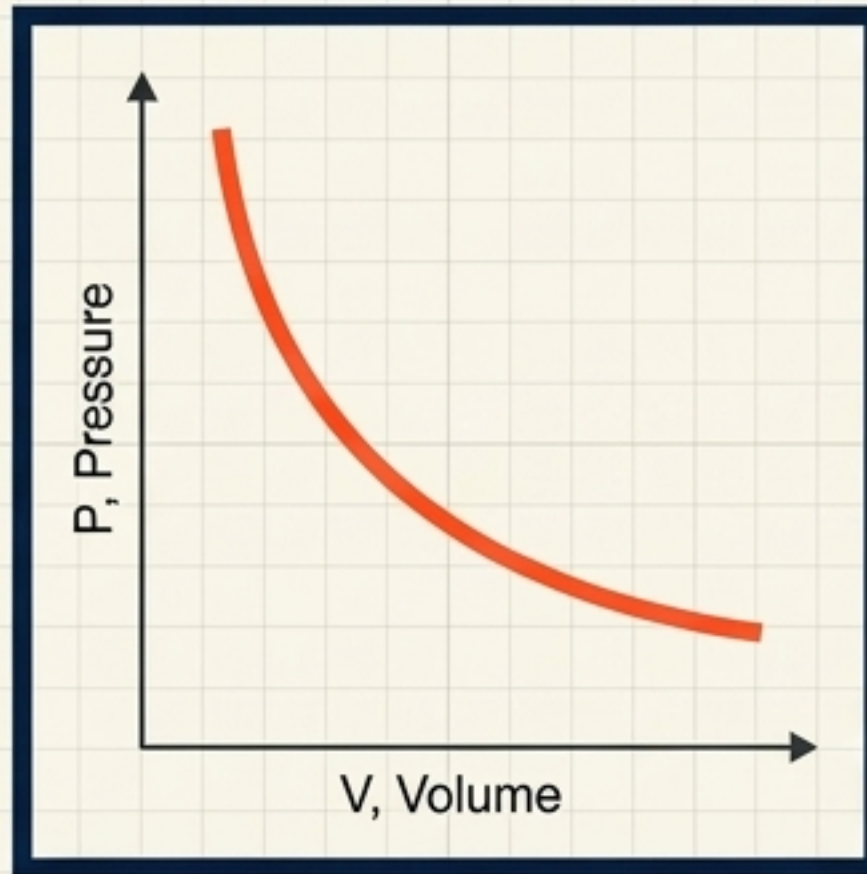
The engineering scale and cost of a compressor are dictated by two exact metrics:

- **Overall Compression Ratio (RT):** The mechanical intensity of the pressure increase.
- **Compression Unit Capacity:** The raw quantity of gas processed in a specific unit of time (measured in SCF/hr or SCF/D).

Compressor classification separates positive displacement from dynamic systems

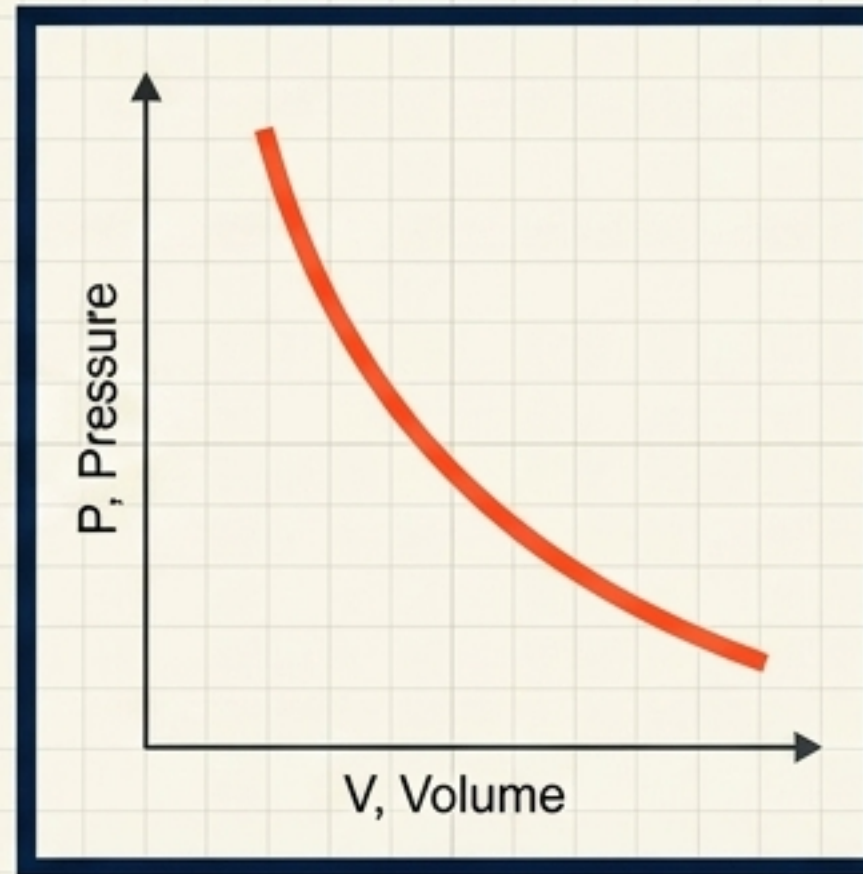


Three thermodynamic pathways define the gas compression process



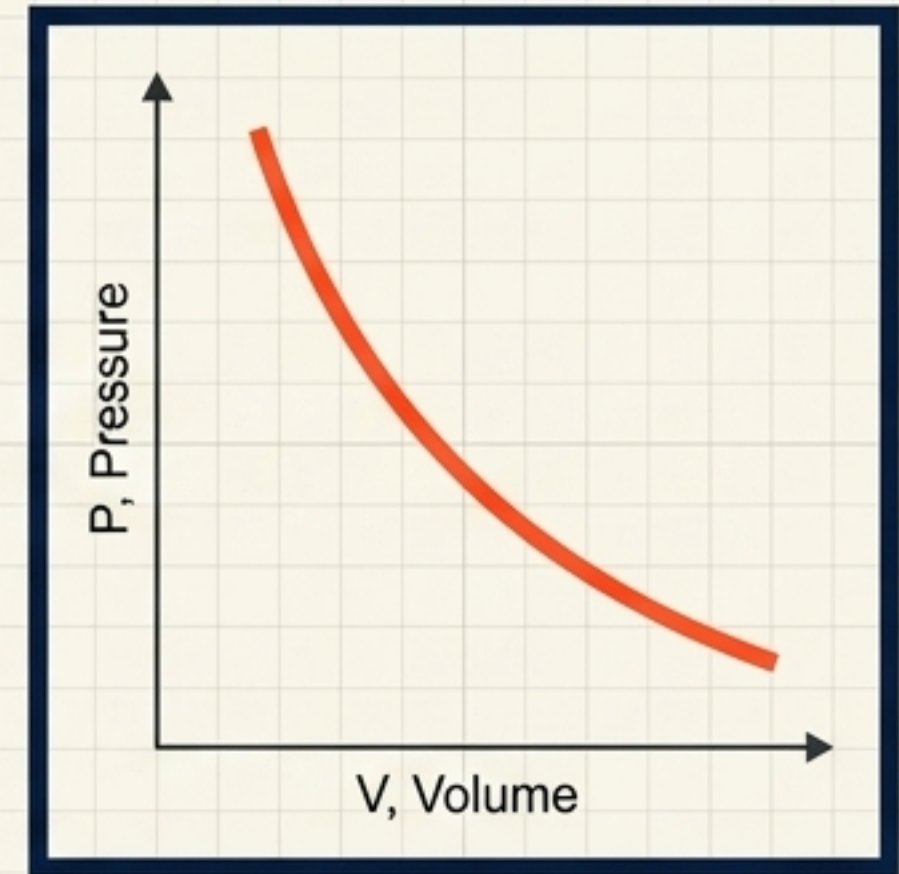
Isothermal Process

The temperature of the gas is mechanically maintained at a constant state throughout compression.



Isentropic Process

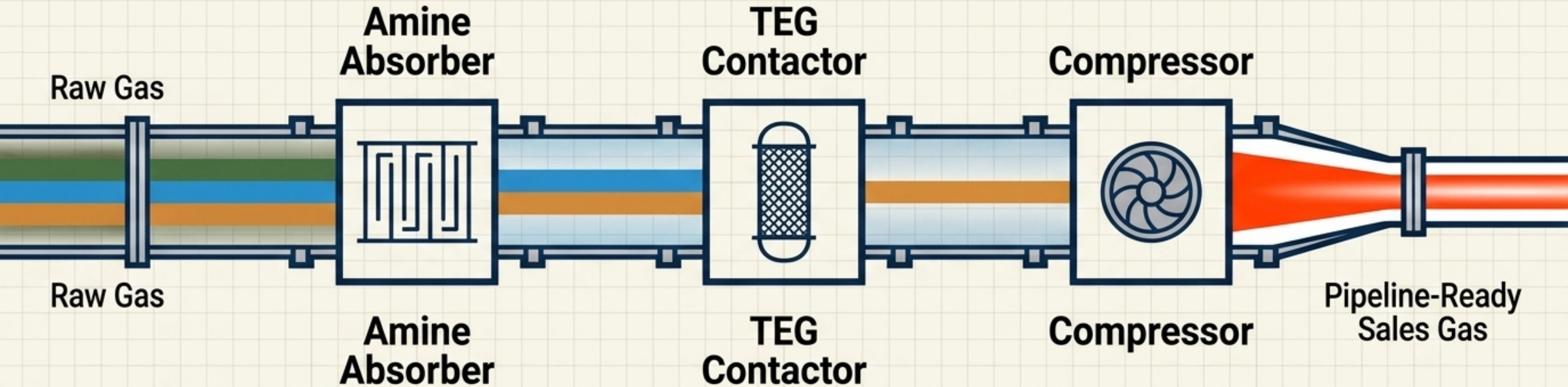
(Constant Entropy) A theoretical, frictionless process where absolutely no heat is added to or removed from the gas during compression.



Polytropic Compression

The real-world operational standard. It behaves similarly to isentropic compression but accounts for actual friction and heat transfer (it is not strictly adiabatic).

The Plant Overview: Synthesizing raw extraction into market-ready sales gas



By sequentially **stripping** acid gases (**Sweetening**), eliminating destructive water vapor (**Dehydration**), and thermodynamically **empowering the flow** (**Compression**), hazardous raw material is successfully **transformed into pipeline-ready energy**.